

When Planning Fails Despite Correct Execution: On Epistemic Calibration for LLM-Based Multi-Agent Systems

Anonymous Authors¹

Abstract

LLM-based multi-agent systems can fail even when planned actions are executed correctly because agents may misjudge their knowledge when evaluating plan feasibility, a phenomenon we term epistemic miscalibration in planning. Unlike execution errors, epistemic miscalibration is latent during planning, as generated plans can remain self-consistent and executable without observable errors; the miscalibration is also dynamic, as new information can alter feasibility assessments, potentially obscuring past miscalibration signals and causing them to recur over time. To address this, we propose the Epistemic Planning Calibration Agentic Workflow (EPC-AW), which assesses whether plans remain supported under varying information conditions rather than directly verifying feasibility. EPC-AW employs Information-consistency-based Plan Selection, selecting plans whose evaluations are stable across agents, together with Consistency-guided Epistemic State Refinement to adapt calibration over time by leveraging past discrepancies to guide future planning. Experiments show that EPC-AW improves system-level success by an average of 9.75%.

1. Introduction

Large language model-based multi-agent systems have become a dominant paradigm for complex decision making, tool use, and long-horizon task execution (Becattini et al., 2025; Sun et al., 2025; He et al., 2025). By decomposing tasks across interacting agents, such systems enable scalable problem solving beyond the capability of single-agent approaches (Becattini et al., 2025; Ronanki, 2025). However, despite their rapid development, these multi-agent systems remain fragile in practice, with failures frequently observed

in real deployments (Cemri et al., 2025; Hammond et al., 2025). Diagnosing and correcting such failures has therefore become an emerging requirement for reliable agentic systems (Liang et al., 2025; Epperson et al., 2025).

Recently, a growing body of work has sought to repair failures in LLM-based multi-agent systems by intervening at different stages of system operation, including post-hoc and online methods (Liang et al., 2025; Ma et al., 2025). Post-hoc approaches (Epperson et al., 2025; Ma et al., 2025) analyze interaction traces after task failure and apply retrospective corrections, while online methods (Liang et al., 2025; Shen et al., 2025) monitor error accumulation during task progression and intervene to prevent errors from escalating into task failure. Across these approaches, failures are predominantly attributed to execution-level faults, such as logically incorrect tool outputs or invalid tool returns (Zhang et al., 2025b; Lu et al., 2025; Reid et al., 2025).

However, practical deployments reveal a distinct class of failures that cannot be explained by execution faults alone. Even when all actions are correctly executed following the prescribed plan, the task may remain unsolved because the plan itself is infeasible with respect to the intended objective. Such failures arise at the planning stage, where the plan encodes a mismatch between intended goals and the actions required to achieve them. For example, a plan may invoke external tools and receive valid responses, yet fail to acquire verifiable evidence for the objective. This occurs when the plan is based on a miscalibrated feasibility assessment, causing the system to repeatedly execute valid actions without ever satisfying the goal.

We characterize this failure mode as *epistemic miscalibration in planning*: a planning agent assigns unwarranted confidence to feasibility assessments, failing to recognize the limits of its knowledge in tasks requiring iterative information acquisition. This phenomenon poses two key challenges. First, epistemic miscalibration is latent. Unlike execution errors, which manifest as observable errors in action realization or inconsistencies in reasoning traces (Zhang et al., 2025c; Cemri et al., 2025), epistemic miscalibration arises from erroneous feasibility assessments that leave no explicit error signals. Consequently, plans affected by epistemic miscalibration can appear coherent and executable, making

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

such failures particularly difficult to detect at the planning stage. Second, epistemic miscalibration is dynamic. As the planning agent continuously acquires new information, its feasibility assessments evolve, which can gradually obscure past miscalibration signals and lead to repeated miscalibration over time. While prior methods leverage dynamic feedback to correct execution-level errors (Ma et al., 2025; Epperson et al., 2025), they rely on observable errors as supervision and therefore do not address plan-level epistemic miscalibration, limiting their ability to sustain calibration under evolving information.

To mitigate epistemic miscalibration, we propose the *Epistemic Planning Calibration Agentic Workflow* (EPC-AW), a planning-centered workflow for multi-agent collaboration. EPC-AW does not attempt to directly verify plan feasibility. Instead, it focuses on whether the plan continues to be supported by the agents under different information conditions. EPC-AW comprises two complementary components. *Information-consistency-based Plan Selection* (IPS) operates within each round and serves as a planning-time diagnostic. Rather than checking whether the plan is judged as feasible, IPS examines whether a plan’s evaluation remains stable across agents that possess different information. Plans whose evaluation varies substantially across heterogeneous information conditions are treated as epistemically fragile, while plans with stable evaluations are selected. *Consistency-guided Epistemic State Refinement* (CESR) operates across rounds by recording discrepancies between the planning agent’s locally selected plans and the IPS-selected plans, interpreting these discrepancies as signals for epistemic miscalibration. These signals are integrated into persistent memory, which constrains subsequent planning and prevents previously observed miscalibration patterns from reoccurring as information accumulates. Together, EPC-AW shifts failure mitigation from execution-time correction to planning-time epistemic calibration. In summary, this paper makes the following contributions:

- We formalize *epistemic miscalibration in planning* as a repair target in LLM-based multi-agent systems, revealing failures arising from miscalibrated planning assessments even under correct execution.
- To address epistemic miscalibration in planning assessments under the absence of observable errors, we propose IPS, which selects plans whose evaluations remain stable across agents operating under heterogeneous information.
- To adapt epistemic calibration under evolving information, we introduce CESR, a memory-driven mechanism that leverages past calibration errors to constrain future planning and suppress persistent misjudgments.
- Extensive experiments on six LLM-based multi-agent benchmarks demonstrate that EPC-AW significantly improves task success, achieving an average 9.75%

increase in system-level success rate.

2. Related Works

2.1. Failure and Repair in LLM-based Multi-Agent Systems

LLM-based multi-agent systems have been increasingly adopted for complex reasoning, tool use, and long-horizon decision making (Becattini et al., 2025; Ronanki, 2025; Li et al., 2024), but they remain vulnerable to failures caused by reasoning errors, coordination breakdowns, and unreliable information use (Dobrovsky et al., 2025; Hammond et al., 2025; Zhang et al., 2025a). To improve system reliability, prior work has proposed mechanisms for diagnosing and repairing failures by analyzing execution traces and agent interactions during or after runtime (Pan et al., 2025; Shen et al., 2025; Epperson et al., 2025; Ma et al., 2025).

Existing approaches differ in when and how corrections are applied. Some methods perform post-hoc or cross-run debugging by inspecting interaction traces or validating failure hypotheses through repeated executions (Epperson et al., 2025; Ma et al., 2025). Others introduce online monitoring and correction, such as rollback and reflection during execution (Liang et al., 2025) or history-conditioned anomaly detection with local repairs (Shen et al., 2025). Despite these differences, most methods focus on correcting incorrect actions or local reasoning errors during execution.

In contrast, this work formulates *epistemic miscalibration in planning* as a distinct repair target in LLM-based multi-agent systems. It reveals that system-level failures can arise from miscalibrated feasibility assessments at planning time, even when all subsequent executions are locally correct.

2.2. Model- and Agent-Level Epistemic Calibration

Epistemic calibration in large language models has been widely studied in the context of overconfidence, and uncertainty estimation (Abbasi Yadkori et al., 2024; Lee et al., 2025; Chhikara, 2025). This line of work treats miscalibration as a property of an individual model or agent, aiming to align expressed confidence with actual knowledge.

Several methods leverage multi-agent structures to enhance calibration through consensus, voting, or verifier agents that critique intermediate outputs (Clark et al., 2025; Wen et al., 2024; Pitre et al., 2025). These approaches rely on LLMs acting as judges, but such first-order judgments are themselves subject to epistemic miscalibration, limiting their ability to diagnose epistemic failures. Another related line of work draws from peer prediction and Bayesian truth inference (Witkowski & Parkes, 2012; Chen et al., 2025), which calibrate reports using incentive signals and repeated feedback under static game-theoretic assumptions.

In contrast, we study epistemic miscalibration as a system-level failure mode in running LLM-based multi-agent systems. Our goal is to repair task failures by intervening during system operation, where there is no additional payoff signals or external supervision. Motivated by peer-based calibration methods, we exploit the stability of agents' evaluations across heterogeneous information and refine epistemic states over time based on persistent cross-agent inconsistencies. Further details are provided in Appendix F.

3. Problem Formulation

3.1. Operation of LLM-based Multi-Agent Systems

An LLM-based multi-agent system is denoted by $\mathcal{M}$ and consists of multiple interacting agents that coordinate through structured messages, shared memory, and external tools (Li et al., 2025; Wu et al., 2024). Given a user query Q , the system aims to produce an answer by iteratively planning, executing tool-mediated actions, and aggregating newly acquired information.

At iteration t , the system maintains an information context $\mathcal{I}^{(t)}$, which aggregates the user query, previously acquired evidence, and interaction history. Conditioned on $\mathcal{I}^{(t)}$, a planning agent generates a plan $\pi^{(t)} = (g^{(t)}, a^{(t)})$, where $g^{(t)}$ denotes an intermediate goal and $a^{(t)}$ specifies an action intended to acquire information relevant to $g^{(t)}$ through available tools, such as web search or code execution. The selected action is executed, yielding an observation or piece of evidence $e^{(t)}$, which is incorporated into the information context via an aggregation operator

$$\mathcal{I}^{(t+1)} = D(\mathcal{I}^{(t)}, \pi^{(t)}, e^{(t)}), \quad (1)$$

where $D(\cdot)$ integrates new evidence into the existing context.

This planning-execution loop continues until a stopping criterion is met. The overall behavior of the system defines a mapping from the user query to a final answer,

$$\hat{Y} = \mathcal{M}(Q), \quad (2)$$

where $\hat{Y}$ denotes the generated answer to the query Q .

3.2. Epistemic Miscalibration in Planning

Epistemic miscalibration in planning refers to a failure of feasibility assessment, in which the agent assigns unwarranted confidence to its judgments and fails to recognize the limits of its knowledge in tasks that require iterative information acquisition.

Let ϕ denote the latent factual situation of the task, which specifies, in principle, the complete information required to justify a plan. This information is not directly observable and must be acquired through iterative interactions.

Given the information available at step t , denoted by $\mathcal{I}^{(t)}$, the planning agent implicitly affirms the feasibility of a plan $\pi^{(t)}$. The subjective feasibility assessment is formulated as

$$J(\pi^{(t)} | \mathcal{I}^{(t)}). \quad (3)$$

This assessment reflects the agent's confidence that executing the action $a^{(t)}$ in the plan can satisfy the current goal $g^{(t)}$ through external tool invocations.

In contrast, the objective feasibility of a plan $\pi^{(t)}$ under the latent factual situation ϕ is characterized by

$$E(\pi^{(t)} | \phi), \quad (4)$$

which indicates whether the plan can, in principle, be justified given the complete underlying information.

Epistemic miscalibration in planning arises when the agent's feasibility assessment formed under partial information, $J(\pi^{(t)} | \mathcal{I}^{(t)})$, is misaligned with the objective feasibility condition $E(\pi^{(t)} | \phi)$ defined under complete information.

3.3. Problem Statement

Given a user query Q , an LLM-based multi-agent system produces a final answer $\hat{Y}$, whose correctness is evaluated against the ground-truth answer Y^* . In practice, even all executions are correct, failures often arise from epistemic miscalibration during the planning phase.

The problem addressed in this work is to design an agentic workflow that mitigates epistemic miscalibration in planning by guiding the system toward plans whose feasibility assessments remain well-aligned with the information available as the system acquires new evidence. By improving epistemic calibration at the planning stage, the workflow aims to reduce downstream errors and increase the likelihood that the final output satisfies $\hat{Y} = Y^*$.

4. EPC-AW: Epistemic Planning Calibration Agentic Workflow

Epistemic miscalibration originates at planning and thus persists even under correct execution and valid tool outputs. Moreover, continual information updates shift feasibility assessments, allowing miscalibration to recur in the subsequent planning. To address these challenges, we propose the *Epistemic Planning Calibration Agentic Workflow* (EPC-AW), which intervenes directly at planning time by separating planning, execution, and diagnosis under heterogeneous information. The overview is shown in Fig. 1, with architectural details in Sec. 4.1. EPC-AW operates both within rounds and across rounds. Within each round, *Information-consistency-based Plan Selection* (IPS) favors plans whose evaluations remain stable across agents with heterogeneous information, mitigating epistemic miscalibration in planning assessments in the absence of observable error signals

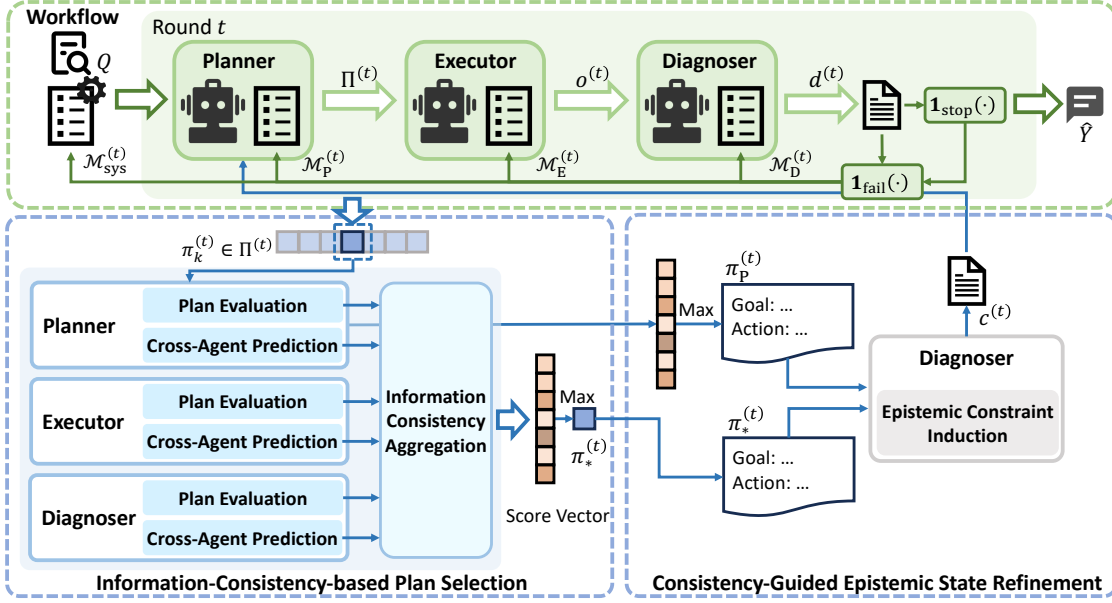


Figure 1. Overview of EPC-AW. EPC-AW consists of three agents, the Planner, Executor, and Diagnoser, each with heterogeneous information in memory. At each round, Information-consistency-based Plan Selection evaluates candidate plans across agents and selects those with stable evaluations, providing a planning-time calibration signal. Across rounds, Consistency-guided Epistemic State Refinement aggregates consistency feedback to guide planning under evolving information. The process terminates upon satisfying a stopping condition, after which the final answer is produced.

(Sec.4.2). Across rounds, *Consistency-guided Epistemic State Refinement* (CESR) aggregates consistency feedback into persistent constraints that guide subsequent planning as information evolves (Sec.4.3).

4.1. System Architecture and Workflow

EPC-AW is instantiated as a role-specialized LLM-based multi-agent system consisting of three agents with fixed roles: a *Planner* (A_P), an *Executor* (A_E), and a *Diagnoser* (A_D). The system operates over discrete interaction rounds $t = 1, 2, \dots, T$. Within each round, agent roles and communication interfaces remain unchanged, forming a stabilized interaction protocol.

Workflow. At round t , the Planner A_P generates an intermediate plan $\pi^{(t)} = (g^{(t)}, a^{(t)})$, where $g^{(t)}$ denotes the goal and $a^{(t)}$ denotes the action intended to acquire evidence relevant to $g^{(t)}$ through external tools. The Executor A_E instantiates $a^{(t)}$ by invoking the corresponding tool with concrete inputs and parameters, producing an execution outcome $o^{(t)}$. The Diagnoser A_D evaluates whether the outcome $o^{(t)}$ meets the plan $\pi^{(t)}$, producing diagnostic feedback $d^{(t)}$ that is communicated back to the Planner.

Memory Structure. EPC-AW maintains a shared *system-level memory* and *agent-level memories* that are private to individual agents. The agent-level memories induce heterogeneous epistemic states across agent roles.

At round t , the complete interaction history is denoted as $\mathcal{H}^{(t)} = \{(\pi^{(k)}, o^{(k)}, d^{(k)})\}_{k=1}^{t-1}$, where $\pi^{(k)}$ is the planned action, $o^{(k)}$ the execution outcome, and $d^{(k)}$ the diagnostic evaluation. This complete history is not directly accessible to any single agent. To formalize the execution analysis within $d^{(k)}$, we define a diagnostic indicator as

$$\mathbf{1}_{\text{fail}}(d^{(k)}) = \begin{cases} 1, & \text{if } d^{(k)} = \text{UNSUPPORTED}, \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

where $d^{(k)} = \text{UNSUPPORTED}$ indicates that the execution outcome $o^{(k)}$ does not meet the plan $\pi^{(k)}$.

System-Level Memory. The system-level memory at round t is defined as

$$\mathcal{M}_{\text{sys}}^{(t)} = \langle Q, \Psi^{(t)}, \mathcal{U} \rangle, \quad (6)$$

where Q denotes the user query, $\Psi^{(t)}$ is a set of verifiable evidence accumulated over rounds, and $\mathcal{U} = \{U_P, U_E, U_D\}$ denotes the abstract description of the role for all agents.

At round k , an evidence item $\psi^{(k)}$ is extracted from the execution outcome $o^{(k)}$ only if the execution outcome meets the plan. Accordingly, the system-level memory is updated in a set-augmentation manner:

$$\Psi^{(k)} = \Psi^{(k-1)} \cup \{\psi^{(k)}\} \quad \text{if } \mathbf{1}_{\text{fail}}(d^{(k)}) = 0, \quad (7)$$

Agent-level Memory. Each agent $\mathcal{A}^i \in \{A_P, A_E, A_D\}$ maintains a private agent-level memory $\mathcal{M}_i^{(t)}$, defined as a role-specific projection of $\mathcal{H}^{(t)}$. The Executor maintains a complete execution trace $\mathcal{M}_E^{(t)} = \mathcal{H}^{(t)}$.

In contrast, the Planner and Diagnoser apply complementary selection criteria. The Planner records plans that are not satisfied by the executions:

$$\mathcal{M}_P^{(t)} = \{(\pi^{(k)}, o^{(k)}, d^{(k)}) \in \mathcal{H}^{(t)} \mid \mathbf{1}_{\text{fail}}(d^{(k)}) = 1\}. \quad (8)$$

Conversely, the Diagnoser selectively records executions whose outcomes satisfy the corresponding plan:

$$\mathcal{M}_D^{(t)} = \{(\pi^{(k)}, o^{(k)}, d^{(k)}) \in \mathcal{H}^{(t)} \mid \mathbf{1}_{\text{fail}}(d^{(k)}) = 0\}. \quad (9)$$

Information State. Each agent A_i operates under a private information state

$$\gamma_i^{(t)} \triangleq (U_i, \mathcal{M}_i^{(t)}), \quad (10)$$

which induces heterogeneous information encodings even under a shared system-level memory $\mathcal{M}_{\text{sys}}^{(t)}$.

4.2. Information-consistency-based Plan Selection

In realistic multi-agent planning scenarios, the feasibility of a plan is typically not verifiable at the planning phase. Moreover, epistemic miscalibration does not manifest as explicit execution errors or observable reasoning failures. To address this challenge, we introduce *Information-consistency-based Plan Selection* (IPS). Rather than attempting to only assess feasibility, IPS uses cross-agent evaluation consistency as a planning-time signal of epistemic support. It prioritizes plans whose evaluations remain stable across agents operating under heterogeneous information states.

4.2.1. PLANNER CANDIDATE PLAN GENERATION

At round t , the Planner A_P generates a finite set of candidate plans conditioned on the shared system-level memory $\mathcal{M}_{\text{sys}}^{(t)}$ and its private information state $\gamma_P^{(t)}$:

$$\Pi^{(t)} = A_P(\mathcal{M}_{\text{sys}}^{(t)}, \gamma_P^{(t)}), \quad (11)$$

where $\Pi^{(t)} = \{\pi_1^{(t)}, \dots, \pi_K^{(t)}\}$, and each $\pi_k^{(t)} \in \Pi^{(t)}$ represents a structurally distinct strategy.

4.2.2. INFORMATION-CONDITIONED PLAN EVALUATION

A common evaluation interface maps information states and candidate plans to real-valued scores:

$$\mathcal{E} : (\gamma_i^{(t)}, \mathcal{M}_{\text{sys}}^{(t)}, \Pi^{(t)}) \longrightarrow \mathbb{R}^K. \quad (12)$$

Each agent computes its evaluation vector

$$\mathbf{e}_i^{(t)} = \mathcal{E}(\gamma_i^{(t)}, \mathcal{M}_{\text{sys}}^{(t)}, \Pi^{(t)}), \quad (13)$$

where $e_i^{(t)}(k) \in \mathbf{e}_i^{(t)}$ reflects agent i 's assessment of plan $\pi_k^{(t)}$ under its information state.

4.2.3. CROSS-AGENT EVALUATION PREDICTION

Under epistemic miscalibration, an agent's direct evaluation is biased by its private information, making self-assigned scores an unreliable indicator of feasibility. We therefore leverage cross-agent evaluations under heterogeneous information as signals. Each agent predicts how candidate plans would be evaluated under alternative information states, and plans whose predicted evaluations remain stable across these perspectives are less sensitive to epistemic variation and thus less prone to epistemic miscalibration.

For each agent $j \neq i$, agent i constructs an approximation of j 's information state

$$\hat{\gamma}_{i \rightarrow j}^{(t)} \triangleq (U_j, \mathcal{M}_i^{(t)}). \quad (14)$$

Then, agent i applies the evaluation interface to obtain prediction scores

$$\hat{e}_{i \rightarrow j}^{(t)}(k) = \mathcal{E}(\hat{\gamma}_{i \rightarrow j}^{(t)}, \mathcal{M}_{\text{sys}}^{(t)}, \Pi^{(t)})_k. \quad (15)$$

Collecting these predictions defines the cross-agent evaluation vector $\hat{\mathbf{e}}_{i \rightarrow -i}^{(t)} = \{\hat{e}_{i \rightarrow j}^{(t)}(k)\}_{j \neq i, k \in [K]}$.

Agent i summarizes the predicted evaluations by averaging over peers:

$$\bar{e}_{-i}^{(t)}(k) = \frac{1}{|\mathcal{A}| - 1} \sum_{j \neq i} \hat{e}_{i \rightarrow j}^{(t)}(k), \quad (16)$$

which represents agent i 's estimate of the plan's expected evaluation under heterogeneous information.

4.2.4. INFORMATION-CONSISTENCY AGGREGATION

To quantify cross-agent information consistency, each agent compares its own plan evaluation with the simulated aggregate evaluation from other agents. The agent-level information-consistency score is defined as

$$s_i^{(t)}(k) = \log(e_i^{(t)}(k)) - \log(\bar{e}_{-i}^{(t)}(k)). \quad (17)$$

Aggregating across all agents yields the plan-level information-consistency score:

$$C_{\text{IPS}}(\pi_k^{(t)}) = \frac{1}{|\mathcal{A}|} \sum_{i \in \mathcal{A}} s_i^{(t)}(k). \quad (18)$$

Plans with higher $C_{\text{IPS}}(\pi_k^{(t)})$ indicate closer alignment between an agent’s evaluation and its prediction of peers, serving as a heuristic measure of cross-agent consistency under heterogeneous information states. Then, the IPS-selected plan $\pi_*^{(t)}$ at round t is formulated as

$$\pi_*^{(t)} = \arg \max_{\pi_k^{(t)} \in \Pi^{(t)}} C_{\text{IPS}}(\pi_k^{(t)}). \quad (19)$$

4.3. Consistency-guided Epistemic State Refinement

While Information-consistency-based Plan Selection (IPS) calibrates planning decisions within a single round, it cannot prevent epistemic miscalibration from recurring over long-horizon interactions. As planning proceeds, the Planner’s internal memory is continually updated, inducing non-stationary plan generation behavior that per-round selection alone cannot correct. We therefore introduce *Consistency-guided Epistemic State Refinement* (CESR), a memory-driven mechanism that accumulates cross-round consistency signals and integrates them into persistent epistemic state constraints, shaping subsequent planning and preventing the reintroduction of previously identified miscalibration.

4.3.1. CROSS-ROUND PLAN DIVERGENCE

Within round t , two plans are distinguished. The Planner selects a plan based on its self-conditioned evaluation,

$$\pi_P^{(t)} = \arg \max_{\pi_k^{(t)} \in \Pi^{(t)}} e_P^{(t)}(k), \quad (20)$$

while IPS selects the information-consistent plan $\pi_*^{(t)}$. A divergence $\pi_P^{(t)} \neq \pi_*^{(t)}$ serves as a diagnostic signal that the Planner’s selected plan is more susceptible to epistemic miscalibration than the plan whose evaluations remain stable under heterogeneous information.

4.3.2. EPISTEMIC CONSTRAINT INDUCTION

Upon detecting such a discrepancy, the Diagnoser generates a lightweight epistemic constraint

$$c^{(t)} = \mathcal{G}(\pi_P^{(t)}, \pi_*^{(t)}, \mathcal{M}_{\text{sys}}^{(t)}), \quad (21)$$

which abstracts the salient information and structural features associated with the potential epistemic miscalibration revealed by the discrepancy.

The constraint is accumulated in the Planner’s memory via

$$\mathcal{M}_P^{(t+1)} = \mathcal{M}_P^{(t)} \cup \{c^{(t)}\}. \quad (22)$$

4.4. Answer Generation and Termination Control

At each round, the Diagnoser determines whether the accumulated system-level information suffices to answer the user

query Q . This decision is captured by a binary indicator. We formalize this with a binary indicator:

$$\mathbf{1}_{\text{stop}}^{(t)}(Q, \Psi^{(t)}) = \begin{cases} 1, & \text{if sufficient evidence,} \\ 0, & \text{otherwise.} \end{cases} \quad (23)$$

If 1, it transitions to answer generation; otherwise, the system proceeds to the next round.

Upon termination, the final answer is produced by

$$Y = \mathcal{G}(Q, \Psi^{(t)}, \mathcal{H}^{(t)}), \quad (24)$$

where $\Psi^{(t)}$ contains accumulated verified evidence and $\mathcal{H}^{(t)}$ summarizes the interaction history.

5. Experiment

5.1. Experimental Setup

5.1.1. DATASET

We evaluate on six benchmarks covering diverse reasoning and search demands. **Bamboogle** (Press et al., 2023) targets compositional multi-step reasoning with minimal external search. **2Wiki** (Ho et al., 2020) emphasizes multi-hop question answering with strong dependence on information retrieval across heterogeneous sources. **HotpotQA** (Yang et al., 2018) requires multi-hop reasoning over Wikipedia with moderate search complexity. **Musique** (Trivedi et al., 2022) stresses sequential reasoning with tightly coupled intermediate inferences. **GAIA** (Mialon et al., 2023) evaluates agentic planning in open-world settings, heavily relying on search and tool use. **MedQA** (Yang et al., 2024) focuses on closed-book clinical reasoning without external retrieval. Details are provided in Appendix A.

5.1.2. BASELINES

We compare EPC-AW with three baselines implemented under the AgentFlow framework (Li et al., 2025), which differ in how epistemic miscalibration is handled during the planning phase. *No-Repair* follows the original AgentFlow setting, generating and executing a single forward plan without any diagnosis or recovery. *Retry*, inspired by metacognitive retry strategies in MAST (Shen et al., 2025), detects epistemic miscalibration at the current planning step and re-generates the plan locally using diagnostic feedback. *Rollback*, inspired by COCO (Liang et al., 2025), performs system-level recovery by rolling back the entire system state to a selected historical step before re-planning. Additional details are provided in Appendix B.

5.1.3. IMPLEMENTATION

All agents in EPC-AW are instantiated using Qwen3-Coder-30B (Yang et al., 2025). The model is deployed on a server

Table 1. Accuracy (%) comparison of different repair methods. Δ denotes the absolute improvement of EPC-AW over the corresponding baseline method on each benchmark. Darker gray indicates larger improvement.

Method	Bamboogle		2Wiki		HotpotQA		Musique		GAIA		MedQA	
	Acc.	Δ	Acc.	Δ	Acc.	Δ	Acc.	Δ	Acc.	Δ	Acc.	Δ
No-Repair (AgentFlow)	48.27	$\uparrow 9.86$	39.00	$\uparrow 12.83$	48.33	$\uparrow 12.67$	9.33	$\uparrow 8.00$	7.09	$\uparrow 8.13$	67.11	$\uparrow 7.00$
Retry	51.47	$\uparrow 6.66$	42.83	$\uparrow 9.00$	51.33	$\uparrow 9.67$	10.67	$\uparrow 6.66$	9.17	$\uparrow 6.05$	71.00	$\uparrow 3.11$
Rollback	<u>52.27</u>	$\uparrow 5.86$	<u>44.83</u>	$\uparrow 7.00$	<u>56.33</u>	$\uparrow 4.67$	<u>14.00</u>	$\uparrow 3.33$	<u>10.76</u>	$\uparrow 4.46$	<u>71.56</u>	$\uparrow 2.55$
EPC-AW	58.13	-	51.83	-	61.00	-	17.33	-	15.22	-	74.11	-

equipped with four NVIDIA RTX 4090 GPUs and served via v_{LLM} for efficient inference. During plan generation, the Planner samples $n = 9$ candidate next-step plans with temperature 0.9 to encourage exploration, while all other generations operate at temperature 0 for deterministic execution. Candidate plans are evaluated using a predefined feasibility metric, yielding scores from 1 to 5. The maximum number of iterations is set to 10. In IPS, plan evaluations and predictions are performed independently for each agent, with outputs obtained via separate LLM calls.

The system interacts with five tools following AgentFlow (Li et al., 2025): a base generator, a Python coder, a Google Search, a Wikipedia Search, and a Web Search. We replace AgentFlow’s Google Search API with a lightweight Chrome-based pipeline retrieving the top-10 results and apply keyword filters to exclude HuggingFace repositories, reducing external dependencies and dataset leakage. For evaluation, GPT-4o is employed as an automatic judge to determine whether predictions match ground-truth answers. All results are averaged over three independent runs to mitigate randomness, following the protocol of AgentFlow (Li et al., 2025). Additional details are provided in Appendix. C.

5.2. Performance Analysis

Table 1 summarizes the accuracy of all methods across six benchmarks. All repair-based methods consistently outperform the No-Repair baseline, indicating that addressing epistemic miscalibration mitigates task failures. On average, Retry improves accuracy from 36.52% to 39.41%, while Rollback further increases performance to 41.63% (+5.11% over No-Repair). EPC-AW achieves the best results on all benchmarks, yielding a 9.75% absolute improvement over No-Repair and a 4.64% gain over Rollback.

Comparing baselines, Retry performs local re-planning at the current step, whereas Rollback enables recovery from earlier planning decisions by reverting the system state. This broader intervention scope allows Rollback to consistently outperform Retry across tasks. However, both Retry and Rollback rely on agent-level diagnosis and analysis. Since LLM-based agents themselves can be epistemically miscalibrated, these diagnostic signals may be unreliable, which

inherently limits the effectiveness of repair.

By contrast, EPC-AW intervenes directly during planning. Through posterior aggregation in IPS and persistent corrective constraints in CESR, EPC-AW reduces epistemic risk during plan selection and prevents recurrent miscalibration across rounds, resulting in substantially stronger and more consistent performance gains.

Table 2. Ablation study on four representative benchmarks, evaluating the individual and combined effects of IPS and CESR.

Method	Bamboogle	2Wiki	HotpotQA	GAIA
EPC-AW [†]	45.87	<u>40.00</u>	<u>51.33</u>	<u>10.24</u>
EPC-AW [‡]	<u>46.67</u>	36.83	49.00	<u>10.24</u>
EPC-AW	58.13	51.83	61.00	15.22

5.3. Ablation Study

We conduct ablation studies on four benchmarks spanning distinct reasoning and search characteristics. GAIA and 2Wiki represent search-dominant settings, with GAIA emphasizing open-world exploration and tool use, and 2Wiki focusing on retrieval-intensive multi-hop QA, whereas Bamboogle targets closed-world compositional reasoning with minimal search and HotpotQA occupies an intermediate reasoning and retrieval regime. The remaining benchmarks are omitted due to substantial overlap with the selected set.

To isolate component contributions, we evaluate two controlled variants. EPC-AW[†] removes both IPS and CESR, approximating planning without epistemic repair with agents under heterogeneous information, while EPC-AW[‡] retains IPS but disables CESR, corresponding to plan selection without constraint feedback.

As shown in Table 2, IPS alone exhibits regime-dependent effects. On the reasoning-centric Bamboogle benchmark, EPC-AW[‡] yields modest gains, indicating that IPS can regularize planning by favoring epistemically supported reasoning paths. However, on search-intensive benchmarks such as 2Wiki, the same variant degrades performance, suggesting that IPS in isolation induces overly conservative planning that suppresses necessary exploration.

CESR converts the conservatism induced by IPS into structured planning guidance. By detecting epistemic miscalibration via plan selection discrepancies and imposing memory-based cross-round constraints, CESR filters unsafe plans without suppressing necessary exploration. As a result, the full EPC-AW consistently outperforms both ablations, indicating that intra-round plan selection and cross-round constraints are jointly essential for robust agentic planning.

Table 3. Comparison between EPC-AW with Mean-Score Aggregation (MSA) and IPS on four representative benchmarks.

Method	Bamboogle	2Wiki	HotpotQA	GAIA
w/ MSA	56.27	52.33	50.67	11.55
w/ IPS	58.13	51.83	61.00	15.22

5.4. Mean-Score Aggregation vs. Cross-Agent Consistency

We compare Information-consistency-based Plan Selection (IPS) with a score aggregation baseline, denoted as Mean-Score Aggregation (MSA), which selects plans by averaging feasibility scores from three agents operating under heterogeneous information. This comparison isolates cross-agent consistency from naive score aggregation, highlighting their distinct effects in agentic planning.

Results in Table 3 reveal a clear task-dependent pattern. On 2Wiki, a predominantly search-oriented benchmark with limited reasoning depth, MSA slightly outperforms IPS (52.33% vs. 51.83%). In this setting, performance is largely driven by information retrieval, and averaging feasibility estimates provides a reasonable approximation.

In contrast, IPS consistently outperforms MSA on tasks requiring multi-step reasoning and exploration. On HotpotQA and GAIA, IPS achieves gains of 10.33% and 3.67%, respectively. These tasks exhibit stronger agent-level disagreement, where naive score averaging masks inconsistencies across reasoning paths and leads to overconfident yet weakly grounded plan selection. By favoring plans that remain consistent across agent assessments, IPS enables EPC-AW to better mitigate miscalibration in complex planning regimes.

5.5. Hyperparameter Sensitivity

We analyze the sensitivity of EPC-AW to the number of sampled plans n in IPS, varying $n \in \{1, 3, 5, 7, 9\}$ across all datasets. When $n = 1$, IPS degenerates to generating a single plan under heterogeneous information, leaving no alternative plans under different knowledge states for comparison. As a result, EPC-AW cannot assess cross-agent consistency and yields the weakest performance.

Once $n > 1$, performance improves markedly, with the largest gain observed from $n = 1$ to $n = 3$, indicating

that the main benefit comes from introducing epistemic diversity rather than exhaustive sampling. As n further increases, performance continues to improve monotonically with diminishing returns, as additional samples provide denser coverage of the epistemic space and enable more reliable epistemic calibration. This consistent trend also suggests that EPC-AW is robust to the choice of n and does not rely on delicate hyperparameter tuning.

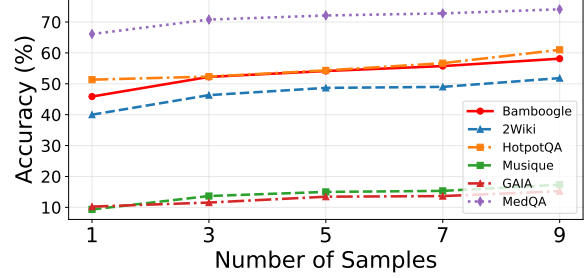


Figure 2. Failure modes in LLM-based multi-agent systems.

5.6. Time Cost

EPC-AW’s runtime is dominated by LLM inference and external tool calls, while internal bookkeeping such as memory updates and score aggregation incurs negligible overhead. Base LLM calls take about 3s, Google Search about 80s due to potential multiple downstream requests, Wikipedia Search about 40s, and other tools (Python coder, generic web search) about 5s per invocation. LLM feedback for planning, execution, and diagnosis consistently takes approximately 3s per call. Compared to Retry and Rollback, which add roughly one extra LLM call per intervention, EPC-AW introduces six additional evaluation and cross-agent prediction calls. These calls are independent and can be executed in parallel, so the time cost is similar to a single LLM invocation. Overall, EPC-AW incurs modest overhead while substantially improving epistemic reliability. Detailed timings are reported in Appendix E.

6. Conclusion

We identify epistemic miscalibration during planning as a distinct and previously underexplored failure mode in LLM-based multi-agent systems. Unlike execution errors, such miscalibration remains latent and dynamic during planning, as plans can appear executable while feasibility assessments evolve with newly acquired information. By shifting mitigation from execution-time correction to planning-time calibration, our approach improves overall system reliability. EPC-AW achieves the best results on all benchmarks, yielding a 9.75% absolute improvement on the system-level accuracy. These results underscore the importance of treating epistemic calibration as a first-class, system-level consideration in the design of robust multi-agent workflows.

7. Impact Statement

This paper presents work whose goal is to improve the correctness and reliability of LLM-based multi-agent systems by mitigating epistemic miscalibration during planning. By reducing unsupported feasibility assessments, the proposed approach improves task success in settings with incomplete and evolving information, and may contribute to more reliable deployment of LLM-based systems in applications where planning quality directly affects outcomes. The work advances system-level planning reliability without introducing new capabilities or ethical concerns beyond those already well studied in large language model deployment.

References

- Abbasi Yadkori, Y., Kuzborskij, I., György, A., and Szepesvari, C. To believe or not to believe your llm: Iterative prompting for estimating epistemic uncertainty. *Advances in Neural Information Processing Systems*, 37:58077–58117, 2024.
- Arif, S., Farid, S., Azeemi, A. H., Athar, A., and Raza, A. A. The fellowship of the llms: Multi-agent workflows for synthetic preference optimization dataset generation. *arXiv preprint arXiv:2408.08688*, 2024.
- Becattini, M., Verdecchia, R., and Vicario, E. Sallma: A software architecture for llm-based multi-agent systems. In *2025 IEEE/ACM International Workshop New Trends in Software Architecture (SATrends)*, pp. 5–8. IEEE, 2025.
- Cemri, M., Pan, M. Z., Yang, S., Agrawal, L. A., Chopra, B., Tiwari, R., Keutzer, K., Parameswaran, A., Klein, D., Ramchandran, K., et al. Why do multi-agent llm systems fail? *arXiv preprint arXiv:2503.13657*, 2025.
- Chen, B., Zhu, T., Han, J., Li, L., Li, G., and Dai, X. Incentivizing truthful language models via peer elicitation games. *arXiv preprint arXiv:2505.13636*, 2025.
- Chhikara, P. Mind the confidence gap: Overconfidence, calibration, and distractor effects in large language models. *arXiv preprint arXiv:2502.11028*, 2025.
- Cho, Y.-M., Guntuku, S. C., and Ungar, L. Herd behavior: Investigating peer influence in llm-based multi-agent systems. *arXiv preprint arXiv:2505.21588*, 2025.
- Clark, N., Shen, H., Howe, B., and Mitra, T. Epistemic alignment: A mediating framework for user-llm knowledge delivery. *arXiv preprint arXiv:2504.01205*, 2025.
- Dobrovsky, A., Schekotihin, K., and Burmer, C. Managing complex failure analysis workflows with llm-based reasoning and acting agents. *arXiv preprint arXiv:2506.15567*, 2025.
- Epperson, W., Bansal, G., Dibia, V. C., Fournay, A., Gerrits, J., Zhu, E., and Amershi, S. Interactive debugging and steering of multi-agent ai systems. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, pp. 1–15, 2025.
- Gong, W., Qiao, S., Cao, C., Tan, S., Ye, J., Liu, H., Chen, S., and Wang, X. Harnessing collective intelligence of multi-agent llm systems for sensor failure reasoning in smart manufacturing. *Journal of Industrial Information Integration*, pp. 101012, 2025.
- Gu, L., Zhu, Y., Sang, H., Wang, Z., Sui, D., Tang, W., Harrison, E., Gao, J., Yu, L., and Ma, L. Medagentaudit: Diagnosing and quantifying collaborative failure modes in medical multi-agent systems. *arXiv preprint arXiv:2510.10185*, 2025.
- Gupta, A. Verifiability-first agents: Provable observability and lightweight audit agents for controlling autonomous llm systems. *arXiv preprint arXiv:2512.17259*, 2025.
- Hagar, N., Agustianto, W., and Diakopoulos, N. Not wrong, but untrue: Llm overconfidence in document-based queries. *arXiv preprint arXiv:2509.25498*, 2025.
- Hammond, L., Chan, A., Clifton, J., Hoelscher-Obermaier, J., Khan, A., McLean, E., Smith, C., Barfuss, W., Foerster, J., Gavenčiak, T., et al. Multi-agent risks from advanced ai. *arXiv preprint arXiv:2502.14143*, 2025.
- He, J., Treude, C., and Lo, D. Llm-based multi-agent systems for software engineering: Literature review, vision, and the road ahead. *ACM Transactions on Software Engineering and Methodology*, 34(5):1–30, 2025.
- Ho, X., Nguyen, A.-K. D., Sugawara, S., and Aizawa, A. Constructing a multi-hop qa dataset for comprehensive evaluation of reasoning steps. In *Proceedings of the 28th International Conference on Computational Linguistics*, pp. 6609–6625, 2020.
- Kim, M., Kim, S., and Thorne, J. From evidence to belief: A bayesian epistemology approach to language models. *arXiv preprint arXiv:2504.19622*, 2025.
- Lee, D., Hwang, Y., Kim, Y., Park, J., and Jung, K. Are llm-judges robust to expressions of uncertainty? investigating the effect of epistemic markers on llm-based evaluation. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 8962–8984, 2025.
- Li, X., Wang, S., Zeng, S., Wu, Y., and Yang, Y. A survey on llm-based multi-agent systems: workflow, infrastructure, and challenges. *Vicinagearth*, 1(1):9, 2024.

- Li, Z., Zhang, H., Han, S., Liu, S., Xie, J., Zhang, Y., Choi, Y., Zou, J., and Lu, P. In-the-flow agentic system optimization for effective planning and tool use. 2025.
- Liang, C., Gan, J., Hong, K., Tian, Q., Wu, Z., and Li, R. Coco: Cognitive operating system with continuous oversight for multi-agent workflow reliability. *arXiv preprint arXiv:2508.13815*, 2025.
- Lin, Y.-C., Chen, K.-C., Li, Z.-Y., Wu, T.-H., Wu, T.-H., Chen, K.-Y., Lee, H.-y., and Chen, Y.-N. Creativity in llm-based multi-agent systems: A survey. *arXiv preprint arXiv:2505.21116*, 2025.
- Liu, X., Chen, T., Da, L., Chen, C., Lin, Z., and Wei, H. Uncertainty quantification and confidence calibration in large language models: A survey. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 2*, pp. 6107–6117, 2025.
- Lu, R., Li, Y., and Huo, Y. Exploring autonomous agents: A closer look at why they fail when completing tasks. *arXiv preprint arXiv:2508.13143*, 2025.
- Ma, M., Zhang, J., Yang, F., Kang, Y., Lin, Q., Rajmohan, S., and Zhang, D. Dover: Intervention-driven auto debugging for llm multi-agent systems. *arXiv preprint arXiv:2512.06749*, 2025.
- Mialon, G., Fourrier, C., Wolf, T., LeCun, Y., and Scialom, T. Gaia: a benchmark for general ai assistants. In *The Twelfth International Conference on Learning Representations*, 2023.
- Nguyen-Hien, T. D., Ivanova, D. R., Teh, Y. W., and Lee, W. S. Extending epistemic uncertainty beyond parameters would assist in designing reliable llms. *arXiv preprint arXiv:2506.07448*, 2025.
- Pan, M. Z., Cemri, M., Agrawal, L. A., Yang, S., Chopra, B., Tiwari, R., Keutzer, K., Parameswaran, A., Ramchandran, K., Klein, D., et al. Why do multiagent systems fail? In *ICLR 2025 Workshop on Building Trust in Language Models and Applications*, 2025.
- Pitre, P., Ramakrishnan, N., and Wang, X. Consensagent: Towards efficient and effective consensus in multi-agent llm interactions through sycophancy mitigation. In *Findings of the Association for Computational Linguistics: ACL 2025*, pp. 22112–22133, 2025.
- Press, O., Zhang, M., Min, S., Schmidt, L., Smith, N. A., and Lewis, M. Measuring and narrowing the compositionality gap in language models. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pp. 5687–5711, 2023.
- Reid, A., O’Callaghan, S., Carroll, L., and Caetano, T. Risk analysis techniques for governed llm-based multi-agent systems. *arXiv preprint arXiv:2508.05687*, 2025.
- Richards, K.-A. Post-cognitive epistemology: Rethinking knowledge, authority, and meaning in the age of predictive language. *Authority, and Meaning in the Age of Predictive Language (July 01, 2025)*, 2025.
- Ronanki, K. Facilitating trustworthy human-agent collaboration in llm-based multi-agent system oriented software engineering. In *Proceedings of the 33rd ACM International Conference on the Foundations of Software Engineering*, pp. 1333–1337, 2025.
- Shen, X., Zhang, Q., Wang, S., Tan, Z., Zhao, X., Yao, L., Tadiparthi, V., Mahjoub, H. N., Pari, E. M., Lee, K., et al. Metacognitive self-correction for multi-agent system via prototype-guided next-execution reconstruction. *arXiv preprint arXiv:2510.14319*, 2025.
- Sun, C., Huang, S., and Pompili, D. Llm-based multi-agent decision-making: Challenges and future directions. *IEEE Robotics and Automation Letters*, 2025.
- Sung, Y. Y., Kim, H., and Zhang, D. Verila: A human-centered evaluation framework for interpretable verification of llm agent failures. *arXiv preprint arXiv:2503.12651*, 2025.
- Tripathi, Y., Kohli, V., and Misra, S. Confident but incorrect: Mitigating hallucination and overconfidence in agentic ai coders. In *2025 Eighth International Conference on Image Information Processing (ICIIP)*, pp. 723–728. IEEE, 2025.
- Trivedi, H., Balasubramanian, N., Khot, T., and Sabharwal, A. Musique: Multihop questions via single-hop question composition. *Transactions of the Association for Computational Linguistics*, 10:539–554, 2022.
- Wan, G., Sun, L., Dou, L., Shi, Z., Wu, F., Jiang, E. H., Huang, W., Zhang, G., Geng, H., Tang, X., et al. Diagnose, localize, align: A full-stack framework for reliable llm multi-agent systems under instruction conflicts. *arXiv preprint arXiv:2509.23188*, 2025.
- Wang, R., Lin, Q., Liu, J., Zong, Q., Zheng, T., Wang, W., and Song, Y. Prospect theory fails for llms: Revealing instability of decision-making under epistemic uncertainty. *arXiv preprint arXiv:2508.08992*, 2025.
- Weaver, R. and Prelec, D. Creating truth-telling incentives with the bayesian truth serum. *Journal of Marketing Research*, 50(3):289–302, 2013.
- Wen, B., Xu, C., Wolfe, R., Wang, L. L., Howe, B., et al. Mitigating overconfidence in large language models: A

behavioral lens on confidence estimation and calibration. In *NeurIPS 2024 Workshop on Behavioral Machine Learning*, 2024.

Wen, C., Liu, J., and Du, L. Consensus planning boosts llm code generation. *Information and Software Technology*, pp. 108030, 2026.

Witkowski, J. and Parkes, D. A robust bayesian truth serum for small populations. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 26, pp. 1492–1498, 2012.

Wu, Q., Bansal, G., Zhang, J., Wu, Y., Li, B., Zhu, E., Jiang, L., Zhang, X., Zhang, S., Liu, J., et al. Autogen: Enabling next-gen llm applications via multi-agent conversations. In *First Conference on Language Modeling*, 2024.

Yang, A., Li, A., Yang, B., Zhang, B., Hui, B., Zheng, B., Yu, B., Gao, C., Huang, C., Lv, C., et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.

Yang, H., Chen, H., Guo, H., Chen, Y., Lin, C.-S., Hu, S., Hu, J., Wu, X., and Wang, X. Llm-medqa: Enhancing medical question answering through case studies in large language models. *arXiv preprint arXiv:2501.05464*, 2024.

Yang, Z., Qi, P., Zhang, S., Bengio, Y., Cohen, W., Salakhutdinov, R., and Manning, C. D. Hotpotqa: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 conference on empirical methods in natural language processing*, pp. 2369–2380, 2018.

Zhang, G., Wang, J., Chen, J., Zhou, W., Wang, K., and Yan, S. Agentracer: Who is inducing failure in the llm agentic systems?, 2025a. URL <https://arxiv.org/abs/2509.03312>.

Zhang, L., Zhai, Y., Jia, T., Huang, X., Duan, C., and Li, Y. Agentfm: Role-aware failure management for distributed databases with llm-driven multi-agents. In *Proceedings of the 33rd ACM International Conference on the Foundations of Software Engineering*, pp. 525–529, 2025b.

Zhang, S., Yin, M., Zhang, J., Liu, J., Han, Z., Zhang, J., Li, B., Wang, C., Wang, H., Chen, Y., et al. Which agent causes task failures and when? on automated failure attribution of llm multi-agent systems. *arXiv preprint arXiv:2505.00212*, 2025c.

Zhou, C., Tang, Y., Chen, K., Bai, X., Qi, S., Shen, L., and Zhang, M. From fragmentation to systematic design: Architecting llm-based multi-agent systems. *Authorea Preprints*, 2026.

A. Dataset Details

We evaluate our approach on a diverse set of reasoning benchmarks that span a broad spectrum of dependence on internal inference versus external information retrieval.

Bamboogle (Press et al., 2023) is a manually constructed multi-step reasoning dataset designed to probe compositional inference over interconnected facts. Each question typically requires up to four explicit inferential steps, with minimal reliance on external search.

2Wiki (Ho et al., 2020) combines structured knowledge from Wikidata with unstructured evidence from Wikipedia, requiring models to retrieve and integrate information across heterogeneous sources. While annotated reasoning chains facilitate interpretable multi-hop inference, successful reasoning is contingent on accurate intermediate retrieval, making the dataset jointly dependent on search quality and sequential reasoning.

HotpotQA (Yang et al., 2018) is a widely used multi-hop question answering benchmark constructed from Wikipedia articles. Questions require integrating information from multiple documents within a relatively homogeneous textual space.

Musique (Trivedi et al., 2022) is a challenging multi-step reasoning dataset characterized by strong sequential dependencies, where each inference stage relies critically on conclusions derived from prior steps.

GAIA (Mialon et al., 2023) is a benchmark designed to assess general AI agents in open-world settings, requiring capabilities such as multi-step reasoning, web navigation, and tool use. Unlike conventional QA datasets, GAIA emphasizes planning and information acquisition under partial observability, with reasoning tightly coupled to search and tool utilization.

MedQA (Yang et al., 2024) consists of multiple-choice questions derived from professional medical licensing examinations. The dataset evaluates clinical reasoning grounded in domain knowledge, requiring models to infer diagnoses or treatments through structured medical reasoning rather than external information retrieval. As such, MedQA primarily tests knowledge-intensive reasoning in a closed-book setting.

Overall, these datasets collectively cover a continuum from reasoning-dominant, closed-world inference to search- and tool-driven open-world problem solving. This diversity allows us to systematically examine how epistemic miscalibration manifests across different planning and reasoning conditions in LLM-based agentic systems.

We follow AgentFlow (Li et al., 2025) and directly adopt its evaluation subsets for all datasets to ensure comparability with prior agent-based reasoning systems. Details are shown in Table 4.

Table 4. Evaluation subsets adopted from AgentFlow. Each dataset occupies a single row, and all samples are uniformly drawn from the official splits and fixed across all methods.

Dataset	Bamboogle	2Wiki	HotpotQA	Musique	GAIA	MedQA
#Samples	125	200	100	200	127	300

B. Baseline Details

This section provides detailed descriptions of the baselines used in our experiments. Notably, neither MAST (Shen et al., 2025) nor COCO (Liang et al., 2025) has released official implementations at the time of writing. Accordingly, all baselines are our re-implementations inspired by the methodological descriptions in the original papers, instantiated within the AgentFlow framework (Li et al., 2025). All methods share the same agent architecture, system memory design, and execution protocol, ensuring a controlled and fair comparison.

Across all baselines, agents maintain access to a shared system memory that records the complete execution history, including generated plans (goals and actions) and corresponding execution results. For baselines involving epistemic assessment, diagnosis is performed exclusively during the planning phase.

No-Repair (AgentFlow). The No-Repair baseline directly follows the original AgentFlow design. The agent generates a complete plan in a single forward pass and executes it without any explicit feedback, diagnosis, or recovery mechanism. This baseline serves as a reference for unmitigated epistemic miscalibration.

Retry (Planning-Level Metacognitive Re-Generation). The Retry baseline adapts metacognitive retry strategies originally proposed in MAST (Shen et al., 2025) to the planning stage. At each planning step, the agent evaluates whether the newly generated plan is epistemically miscalibrated based on the execution history up to the current step. If miscalibration is detected, the agent re-generates the current planning step while preserving all prior history. The diagnostic analysis is explicitly provided as feedback to guide the re-generation. Importantly, Retry does not modify the global system state or historical memory; it only replaces the current planning output.

Rollback (System-Level State Recovery). The Rollback baseline is inspired by the contextual rollback mechanism in COCO (Liang et al., 2025). When epistemic miscalibration is detected at the current planning step, the agent first determines an appropriate historical step to revert to. The entire system state, including shared memory and recorded execution history, is rolled back to the selected step. Planning is then re-executed from that point onward, with the diagnostic analysis provided as feedback to guide re-planning. Compared to Retry, Rollback performs a global state recovery rather than a local re-generation, allowing the system to correct errors that may have propagated across multiple planning steps.

Notably, Retry and Rollback differ fundamentally in their scope of intervention. Retry performs a local correction at the current planning step without altering historical state, while Rollback restores the system to a previous state and re-optimizes subsequent planning decisions. Both baselines operate under identical detection criteria and planning-phase constraints, enabling controlled evaluation of their effectiveness in mitigating epistemic miscalibration.

C. Implementation Details

This section provides additional implementation details to support reproducibility and clarify key design choices.

Backbone Model. All agents in EPC-AW, including the Planner, Executor, and Diagnoser, are instantiated using Qwen3-Coder-30B (Yang et al., 2025). The model is deployed on a server equipped with four NVIDIA RTX 4090 GPUs and served via vLLM for efficient inference. This backbone is selected to accommodate the long-context requirements of multi-step agentic planning and execution, where intermediate plans, tool interactions, and execution traces must be preserved across extended reasoning horizons. To ensure a fair comparison, the same backbone model and deployment configuration are used consistently across EPC-AW and all baseline methods.

Planning and Generation Settings. During the planning phase, the Planner generates a fixed number of $n = 9$ candidate next-step plans. To encourage exploration of alternative planning hypotheses, candidate generation is performed with temperature 0.9. All subsequent generations, including execution, diagnosis, and tool-based reasoning, are performed with temperature 0 to ensure deterministic behavior and reduce variance across runs. The maximum number of iterations is set to 10 for all experiments.

In IPS, multi-agent plan evaluation and belief prediction are performed independently and in parallel. Candidate plans are evaluated using a predefined feasibility metric, where agents assign integer scores from 1 to 5 reflecting their epistemic assessment of plan feasibility.

Feasibility Metric

Score 5 — Exceptional Feasibility

The plan is internally coherent, precise, and strongly justified.

- Tool selection and parameters are fully specified and sufficient *in principle* to support the stated sub-goal under the given context.
- Reasoning is complete, logically tight, and optimally grounded in the available information.
- No implicit assumptions or missing steps are required to interpret the plan.

Score 4 — Near-Perfect Feasibility

The plan is coherent and well-aligned with the stated sub-goal.

- Tool selection is correct; parameters are appropriate but may allow minor refinement.

- Reasoning is sound, though some details could be made more explicit.
- The plan is interpretable without major inference.

Score 3 — Strong Feasibility

The plan is plausible and directly addresses the sub-goal.

- Tool selection is mostly correct; some parameters or steps require mild inference.
- Reasoning is generally sound but partially underspecified.
- The plan remains interpretable, though not maximally precise.

Score 2 — Mostly Feasible

The plan is relevant but exhibits notable epistemic gaps.

- Tool selection is reasonable, but parameters are under-specified or ambiguous.
- Reasoning relies on implicit assumptions or missing details.
- Additional clarification would be required for confident interpretation.

Score 1 — Weak Feasibility

The plan shows limited coherence or weak alignment with the sub-goal.

- Tool selection or parameter specification is incomplete or mismatched.
- Reasoning is vague, fragmented, or poorly grounded in the given context.
- The intended effect of the plan is epistemically unclear.

Tool Configuration. Following AgentFlow (Li et al., 2025), the system interacts with five tools: 1) a base generator for default reasoning, 2) a Python coder that generates and executes Python code and returns execution results, 3) Google Search for retrieving relevant web pages, 4) Wikipedia Search for structured knowledge retrieval, and 5) Web Search for page-level information extraction and summarization.

Compared to the original AgentFlow implementation, we replace direct Google Search API calls with a lightweight local retrieval pipeline that collects the top-10 relevant web pages via a local Chrome-based Google Search interface. This modification avoids potential interference from backend LLMs (e.g., Gemini-based models) implicitly involved in official API responses, ensuring that information retrieval does not benefit from stronger external language models.

To further mitigate information leakage due to publicly available datasets, we apply keyword-based filtering during the search stage to exclude HuggingFace repositories and dataset-specific pages. Retrieved pages are then processed by the Web Search tool, followed by LLM-based summarization.

Evaluation Protocol. For evaluation, GPT-4o is employed as an automatic judge to determine whether model predictions match the corresponding ground-truth answers, following standard practice in tool-augmented reasoning benchmarks (Li et al., 2025; Arif et al., 2024). To reduce the impact of stochasticity, all experiments are repeated three times, and we report the average accuracy across runs. All evaluation settings are aligned with those used in AgentFlow to facilitate direct comparison.

D. Appendix: EPC-AW Pseudocode and Explanation

The EPC-AW algorithm iteratively performs planning, execution, and diagnosis in a multi-agent workflow to improve epistemic calibration. The workflow can be described as follows:

- **Initialization:** System memory $\mathcal{M}_{\text{sys}}$ stores the query Q , execution history, and the abstract description of the role for all agents $\mathcal{U}$. Each agent (Planner P , Executor E , Diagnoser D) maintains a private memory $\mathcal{M}_i$. The evidence set Ψ

Algorithm 1 EPC-AW: Epistemic Planning Calibration Agentic Workflow

Input: query Q , maximum rounds T
Output: final answer Y
 Initialize system memory $\mathcal{M}_{\text{sys}} \leftarrow \langle Q, \emptyset, \mathcal{U} \rangle$
 Initialize agent memories $\mathcal{M}_P \leftarrow \emptyset, \mathcal{M}_E \leftarrow \emptyset, \mathcal{M}_D \leftarrow \emptyset$
 Initialize evidence set $\Psi \leftarrow \emptyset$
for $t = 1$ **to** T **do**
 Generate candidate plans $\Pi \leftarrow A_P(\mathcal{M}_{\text{sys}}, \gamma_P)$
 Select plan $\pi^* \leftarrow \text{IPS}(\Pi, \mathcal{M}_{\text{sys}}, \{\gamma_i\})$
 Execute plan and observe outcome $o \leftarrow A_E(\pi^*)$
 Diagnose execution $d \leftarrow A_D(\pi^*, o)$
 if $\mathbf{1}_{\text{fail}}(d) = 0$ **then**
 Extract evidence $\psi \leftarrow \mathcal{F}(d)$
 $\Psi \leftarrow \Psi \cup \{\psi\}$
 end if
 Compute planner-selected plan $\pi_P \leftarrow \arg \max_{\pi_k \in \Pi} e_P(k)$
 Update planner memory $\mathcal{M}_P \leftarrow \text{CESR}(\pi_P, \pi^*, \mathcal{M}_{\text{sys}}, \mathcal{M}_P)$
 if $\mathbf{1}_{\text{stop}}(Q, \Psi) = 1$ **then**
 return $\mathcal{G}(Q, \Psi, \mathcal{H})$
 end if
end for
return $\mathcal{G}(Q, \Psi, \mathcal{H})$

is initialized empty.

- **Candidate Plan Generation:** The Planner generates a set of candidate plans Π based on the system memory and its policy γ_P .
- **Plan Selection:** IPS chooses the most epistemically promising plan π^* from Π , taking into account cross-agent alignment.
- **Plan Execution and Diagnosis:** The Executor executes π^* and observes outcome o . The Diagnoser evaluates the execution outcome o whether it satisfies the plan π^* .
- **Evidence Extraction:** If the execution satisfies the plan π^* ($\mathbf{1}_{\text{fail}}(d) = 0$), evidence ψ is extracted and added to Ψ .
- **Planner Memory Update:** The Planner identifies its selected plan π_P and updates its memory $\mathcal{M}_P$ via CESR, aligning plan selection with accumulated feedback.
- **Termination Condition:** The algorithm stops if the accumulated evidence Ψ suffices to answer query Q ($\mathbf{1}_{\text{stop}}(Q, \Psi) = 1$), returning the final answer $\mathcal{G}(Q, \Psi, \mathcal{H})$.
- **Iteration:** If termination criteria are not met, the loop continues until maximum rounds T are reached.

This workflow ensures that the multi-agent system iteratively refines plans and updates epistemic states, leading to more reliable final answers under uncertainty.

E. Time Cost Analysis

We analyze the time cost of EPC-AW by focusing on external tool invocations and LLM inference time, as these dominate the overall runtime in agentic workflows. Details are shown in Table 5. Moreover, internal bookkeeping operations, such as memory updates and score aggregation, incur negligible overhead and are therefore omitted.

Tool Invocation Cost. The primary source of time consumption arises from external tools. Specifically, a single invocation of the base LLM generator requires approximately 3 seconds on average. The Google Search tool incurs the highest cost, with an average latency of approximately 80 seconds per call, as it may trigger multiple downstream web search requests depending on the retrieved webpages. Similarly, Wikipedia search requires around 40 seconds on average, with the exact latency determined by the number of relevant pages involved and corresponding web search calls. Other tools, including the Python coder and generic web search, incur relatively modest costs of approximately 5 seconds per invocation.

LLM Inference Cost. LLM feedback for planning, execution, and diagnosis typically completes within approximately 3 seconds per call. This latency is stable across tasks and methods due to the use of a fixed backbone model and deterministic decoding for non-planning stages.

Cost of Retry, Rollback, and EPC-AW. The computational overhead of Retry and Rollback mechanisms can be approximated as a single additional LLM invocation per intervention step. In contrast, EPC-AW introduces additional evaluation and belief prediction stages, corresponding to six extra LLM calls in total. However, these calls are independent and can be executed in parallel. With asynchronous inference, their wall-clock latency is therefore comparable to that of a single LLM invocation. As a result, EPC-AW incurs limited additional time overhead while providing substantially improved epistemic calibration.

Table 5. Time Cost Breakdown of EPC-AW Components

Component / Tool	Average Latency (s)
Base LLM generator	3
Google Search	80
Wikipedia Search	40
Python Coder / Web Search	5
LLM Feedback (planning / execution / diagnosis)	3
Retry / Rollback (within one round)	3
EPC-AW (within one round)	Slightly above 3 (parallelizable)

F. Related Work

F.1. Failure and Repair in LLM-based Multi-agent Systems

LLM-based multi-agent systems have been widely adopted for complex decision making, tool use, and long-horizon task execution (Zhou et al., 2026; Ronanki, 2025; Lin et al., 2025). Despite their capabilities, these systems remain prone to failures arising from reasoning errors, coordination breakdowns, and unreliable or inconsistent use of information (Cemri et al., 2025; Hammond et al., 2025; Zhang et al., 2025a). Such failures can propagate across agents and rounds, often compounding errors in long-horizon tasks and multi-step planning scenarios. As a result, a growing body of work (Wan et al., 2025; Gu et al., 2025; Gong et al., 2025) has studied how to diagnose, localize, and repair failures in these systems. Most existing approaches focus on improving reliability by analyzing execution traces, monitoring agent interactions, or applying interventions at different stages of system runtime (Liang et al., 2025; Shen et al., 2025; Epperson et al., 2025; Ma et al., 2025), yet challenges remain in achieving robust cross-agent consistency and epistemic calibration, particularly when agents hold heterogeneous or partial information.

Several methods have been proposed to address failures in LLM-based multi-agent systems, broadly falling into post-hoc or cross-run debugging, and online correction during execution. In the post-hoc category, AGDebugger (Epperson et al., 2025) allows developers to inspect and edit multi-agent interaction traces after execution, enabling identification and manual correction of failures. DoVer (Ma et al., 2025) further formalizes this approach by testing failure hypotheses through targeted interventions and validating their effects across repeated executions, effectively providing a controlled mechanism for debugging and improving system robustness over multiple runs.

In contrast, online correction methods aim to detect and mitigate failures as workflows unfold. COCO (Liang et al., 2025) applies continuous monitoring, rollback, and reflective reasoning to correct execution drift and resolve inter-agent inconsistencies in real time. Similarly, MASC (Shen et al., 2025) leverages history-conditioned anomaly signals to detect problematic execution steps and applies local corrections to prevent cascading errors that could compromise long-horizon tasks. While these methods improve reliability during runtime, they typically operate on observed traces or local signals without explicitly modeling cross-agent epistemic miscalibration, leaving challenges in ensuring robust coordination and consistent belief updating across heterogeneous agents.

Despite differences in how and when repair is applied, these methods mainly target incorrect actions or local reasoning errors but neglect the plan’s commitments, allowing unreliable feasibility assessments to persist. In contrast, this work formulates *epistemic miscalibration in planning* as a distinct repair target in LLM-based multi-agent systems. It reveals

that system-level failures can arise from miscalibrated feasibility assessments at planning time, even when all subsequent executions are locally correct.

F.2. Model- and Agent-Level Epistemic Calibration

A growing body of work studies epistemic calibration in large language models (LLMs), focusing on overconfidence (Wen et al., 2024; Hagar et al., 2025; Tripathi et al., 2025) and uncertainty estimation (Liu et al., 2025; Wang et al., 2025; Lee et al., 2025; Nguyen-Hien et al., 2025). This line of research primarily treats epistemic miscalibration as a property of an individual model or agent, aiming to align a model’s expressed confidence with its actual knowledge. Methods include post-hoc confidence adjustment, uncertainty-aware decoding, and calibration-aware fine-tuning, which have been shown to reduce overconfident errors and improve the reliability of single-agent predictions. Even when multiple agents are considered, the focus largely remains on the calibration of individual LLM judgments rather than on the interactions or coordination of a deployed multi-agent system.

Several approaches leverage multi-agent structures as a means to improve agent-level calibration (Clark et al., 2025; Wen et al., 2024). Consensus or voting mechanisms aggregate agents’ assessments to stabilize judgments (Pitre et al., 2025; Wen et al., 2026), while verifier agents critique intermediate outputs based on detectable errors or invalid responses (Sung et al., 2025; Gupta, 2025). These methods rely on LLMs acting as judges to evaluate content quality, but such first-order judgments are themselves subject to epistemic miscalibration, which can propagate errors and limit their ability to reliably diagnose failures. Additionally, aggregation-based approaches may smooth out but not eliminate systemic biases shared across agents, leaving residual miscalibration unaddressed.

Another related line of work draws from peer prediction and game-theoretic truth inference (Cho et al., 2025; Richards, 2025; Kim et al., 2025). Bayesian Truth Serum (Witkowski & Parkes, 2012; Weaver & Prelec, 2013) and its extensions elicit truthful reporting by rewarding agreement with peers’ predictions. Recent adaptations implement explicit scoring rules or discrimination games for LLM-based agents (Chen et al., 2025), aiming to incentivize accurate and self-consistent outputs. While effective under static Bayesian game assumptions, these approaches rely on repeated interactions, payoff signals, or structured incentives, and thus do not naturally extend to dynamic multi-agent systems with heterogeneous knowledge, limited interventions, and long-horizon task dependencies.

In contrast, we study epistemic miscalibration as a source of failure in LLM-based multi-agent systems. Our goal is to repair task failures by intervening during system operation, where there is no additional payoff signals or external supervision. Inspired by peer-based calibration methods, we instead exploit the stability of agents’ evaluations across heterogeneous information and refine epistemic states over time based on persistent cross-agent inconsistencies.